



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-2027
Course Handout (Part - II- B)

01.08.2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : PHA G619
Course Title : Screening Methods and Techniques in Pharmacology
Instructor-in-Charge : Richa Shrivastava
Team of Instructors : Jagdish Chand, Talati Yugant Krishna, Jyoti Verma

1. Scope and Objective of the Course :

The objective of this course is to understand the principles of experimental pharmacology with special emphasis on *in-vitro* cell culture techniques and *in vivo* experimentation. Basics of animal handling, breeding, diet manipulation, dose Calculation-Human equivalent dose, development of new animal models and evaluation techniques for various diseases and their standardization will be covered. The course also includes toxicological, teratogenic, carcinogenic studies, and data analysis

2. Text Books (TB):

N. S. Parmar, Shiv Prakash, "Screening Methods in Pharmacology"- Alpha Science International, 2006 , Reprint 2011-Narosa publications

3. Reference Books (RB):

1. H Gerhard Vogel, " Drug Discovery and Evaluation- Pharmacological Assays" – II Edition, Springer, 2002
2. Laurence L Brunton, "Goodman and Gilman's Pharmacological Basis of Therapeutics"- XII Edition, Mc-Graw Hill, New York, 2011



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4. Course Plan:

Module No	Lecture session	Ref	Learning Outcome
1: Introduction, Drug Discovery Process	L 1-3: Definition, Scope and common terms used Strategies in Drug discovery and evaluation	RB:2; Class notes	Understanding of the newer drug discovery strategies.
2: General introduction of Animals used for research	L 4-8: Animal handling, breeding and diet manipulation for testing, methods involved Laboratory Animals , Dose calculation-Human equivalent dose	Ch-T-3-4 Class notes	Learning of CPCSEA guidelines, animals handling, breeding, diet requirement and advantage /disadvantage of specific Lab animals/species used for research
3: Introduction of Screening Methods for New chemical Entities/drugs	Evaluation of Drugs Acting on : L: 12-16: a) Central Nervous System Disorder (Anxiety, Depression, memory deficit/AD, PD, Brain Stroke and epilepsy, anxiety) L: 17-22: b) CVS Disorder: (Hypertension, arrhythmia, I/R -injury, hypertrophy etc) L : 23-27: c) Respiratory System (Asthma and COPD) d) Gastric/peptic ulcer Ulcerative Colitis e) Evaluation of Analgesics ,Anti-inflammatory and Antipyretic Drugs f) Evaluation of Drugs acting on CTZ, diabetes, obesity	Ch-T- (5,7,8,9,11, 13) R 1: chapter A,C,D,J,K	Understand the pathology and preclinical testing, evaluation/ methodologies for new chemical entities/ clinical drugs for various disorders/diseases



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4: Drug resistance and pharmacological intervention	L 27-30: Approached to study development of resistance, and cancer	Class Notes	Study and understand the principle of drug resistance development, Newer approaches/methods to manage resistance, In vitro and In Vivo models
5: Introduction to Toxicity studies	L 31-35: a) OECD Guidelines b) Acute oral toxicity studies c) Teratogenicity, Carcinogenicity studies (Annexure III)	RB-2: Class Notes	Understand the guidelines used in toxicological studies
6: Introduction to Pharmacogenomics /genetics	L 8-11: Pharmacogenetics, Stem cell and Gene Therapy, Genetic manipulation of animals i.e knock in/knock down	Class notes	Understanding of concept of transgenic animals, and recent development in the treatment of hereditary diseases.



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List of experiments*:

S. No.	Experiment
1	Introduction to common laboratory animals and their basic handling techniques
2	To study the various routes of drug administration in laboratory animals.
3	To study different blood collection methods in small laboratory animals.
4	To study different animal models for epilepsy.
5	Behavioral methods used in the study of learning and memory.
6	Evaluation of locomotor activity. (Anxiety and Depression)
7	To study <i>in-vivo</i> methods for Anti-inflammatory activity.
8	To study methods to induce experimental Diabetes mellitus, measurement of blood glucose level, and antidiabetic activity.
9	To study methods to induce experimental hypertension and screen anti-hypertensive activity of drugs.
10	To study methods to induce cerebral stroke in different animal models.
11	To study the <i>in-vitro</i> assays for anticancer drug development.

**subject to IAEC approval*

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	25	<TEST_1>	Closed Book
Comprehensive Examination	3 h	45	<TEST_C>	Closed or/and Open Book
Quizzes		10	Continuous	Closed book
Lab component		20	Continuous	Closed or/and Open Book

Laboratory component will be marked on the basis of viva-voce, day to day work, borderline cases at the time of final grading as well as make-up's.

It is not necessary that all the grades (i.e. A to E) would be awarded.



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In borderline cases subjective judgment will be exercised for pull-up's (max. 2%). Basic guiding factors will be regularity, consistency in performance (above average) or/and steady improvement throughout the semester. Students scoring less than 20% will be considered NOT exposed to the subject.

8. Chamber Consultation Hour: 5:00 PM-6:00 PM [Email: richa.shrivastava@pilani.bits-pilani.ac.in]

9. Notices: Notices pertaining to this course will be sent via emails.

10. Make-up Policy: Make-Ups are not given as a routine. It is solely dependent upon the genuineness of the circumstances under which a student fails to appear in a scheduled evaluation component. In such circumstances, prior permission should be obtained from the Instructor-in-Charge. The decision of the Instructor-in-Charge in the above matter will be final

Instructor - in -Charge
PHA G619



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